

2.3 Differentiation

Further Pure 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Implicit differentiation treats y as a function of x ; derivatives of y terms acquire dy/dx . Parametric differentiation uses $dy/dx = (dy/dt)/(dx/dt)$, provided $dx/dt \neq 0$. For polar curves, compute derivatives through $x = r \cos \theta$ and $y = r \sin \theta$. Higher derivatives locate turning-point type and curvature.

Worked example: follow the reasoning

For $x = t^2$ and $y = t^3$, $dy/dx = (3t^2)/(2t) = 3t/2$ for $t \neq 0$.

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Find dy/dx on $x^2 + y^2 = 25$ at $(3,4)$ and state tangent gradient.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Differentiate: $2x+2y$ $y'=0$ [1]; $y'=-x/y$ [1]; at (3,4), $-3/4$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.